

ECE 330 Exam 1: Fall 2021 NAME _____
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Section (Check one) MWF 10am _____ MWF 12pm _____ MWF 2pm _____

1. _____/25

2. _____/25

3. _____/25

4. _____/25

TOTAL _____/100

USEFUL INFORMATION

$$\sin(x) = \cos(x - 90^\circ)$$

$$0 < \theta < 180^\circ \text{ (lag)}$$

$$-180^\circ < \theta < 0 \text{ (lead)}$$

$$\bar{V} = \bar{Z} \bar{I} \bar{S} = \bar{V} \bar{I}^* = P + jQ$$

$$I_L = \sqrt{3} I_\phi \text{ (delta)}$$

$$V_L = \sqrt{3} V_\phi \text{ (wye)}$$

$$\bar{S}_{3\phi} = \sqrt{3} V_L I_L \angle \theta$$

$$\bar{Z}_Y = \bar{Z}_\Delta / 3$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

ABC phase sequence has A at 0, B at -120°, and C at +120°

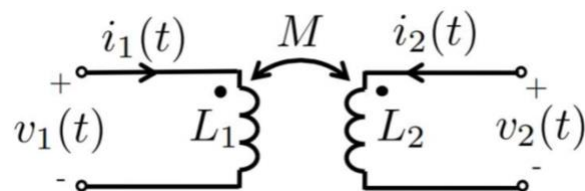
$$\int \underline{H} \cdot \underline{dl} = \int \underline{J_f} \cdot \underline{\hat{n}} dA \quad \int \underline{E} \cdot \underline{dl} = -\frac{d}{dt} \left(\int \underline{B} \cdot \underline{\hat{n}} dA \right) \quad \mathcal{R} = \frac{l}{\mu A} \quad Ni = \mathcal{R} \phi$$

$$\phi = BA$$

$$\lambda = N\phi = Li \text{ (if linear)}$$

$$v = \frac{d\lambda}{dt} k = \frac{M}{\sqrt{L_1 L_2}}$$

$$1 \text{ hp} = 746 \text{ W}$$

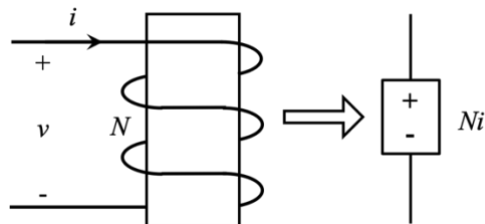
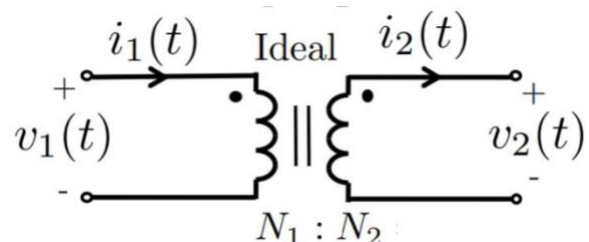


$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

$$a = \frac{N_1}{N_2} \quad N_1 i_1 = N_2 i_2$$

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$



Problem 1 (25 points)

A feeder with an impedance of $0.1+j0.2 \, \Omega$ supplies a single-phase 10 kW 0.8 lagging power factor load. The voltage across the load is $v_L(t) = \sqrt{2} \sin(377t + 40^\circ)$ V. Calculate:

a) The source current phasor $\bar{I}_s$

$\bar{I}_s =$ _____

b) The source (sending end) voltage phasor $\bar{V}_s$

$\bar{V}_s =$ _____

c) The total complex power supplied by the source, $\bar{S}_t$

$\bar{S}_t =$ _____

d) The power factor at the source (sending end). Specify whether it is lagging or leading.

PF = _____

Circle one: Lagging / Leading

Problem 2 (25 points)

Two 3-phase loads are connected in parallel to a wye-connected source of 208V line to line. The loads are given as:

Load 1: wye-connected, 10 A (line current), 3200 W (3-phase), lagging power factor

Load 2: delta-connected, 8 A (phase current), 0.9 power factor lagging.

a) What is the total line current $\bar{I}_L$ being supplied by the source?

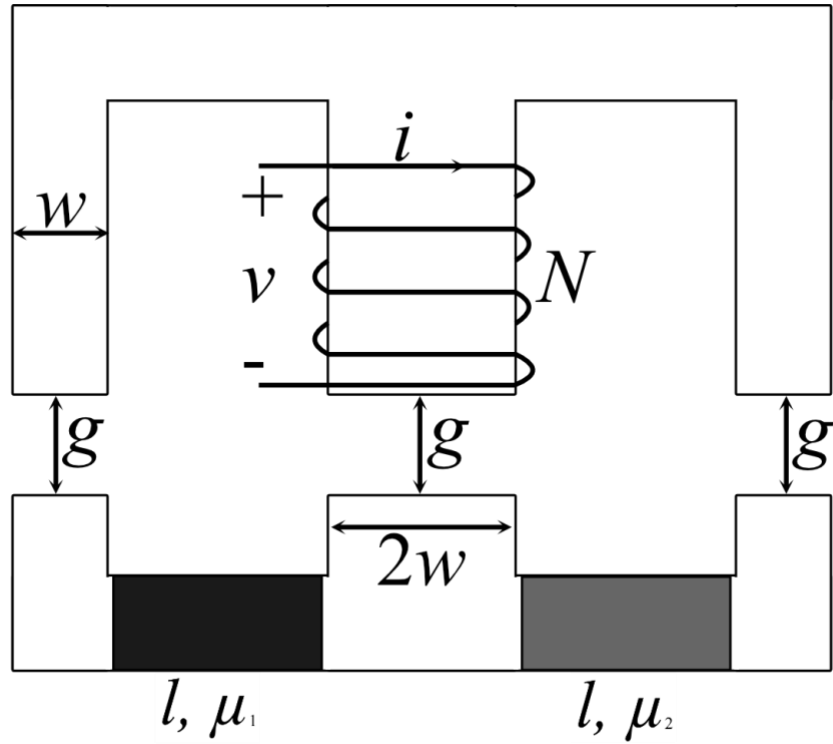
$\bar{I}_L =$ _____

b) How many capacitive VARs PER-PHASE needed to bring overall power factor to unity?
Justify your answer. Answers without justification will receive no credit.

$Q_c =$ _____

Reason:

Problem 3 (25 points)



An iron core with infinite permeability, width $w=2$ cm, air gap length $g=3$ mm, and depth into the page $d=3$ cm has a coil wrapped around it as given above with $N=250$. The iron core also has two inserts in it with length $l=10$ cm and permeabilities $\mu_1 = 2500\mu_0$ and $\mu_2 = 1250\mu_0$. Fringing effects can be neglected.

a) Draw the magnetic equivalent circuit for this system. (8 points)

b) What is the flux through the coil? (12 points)

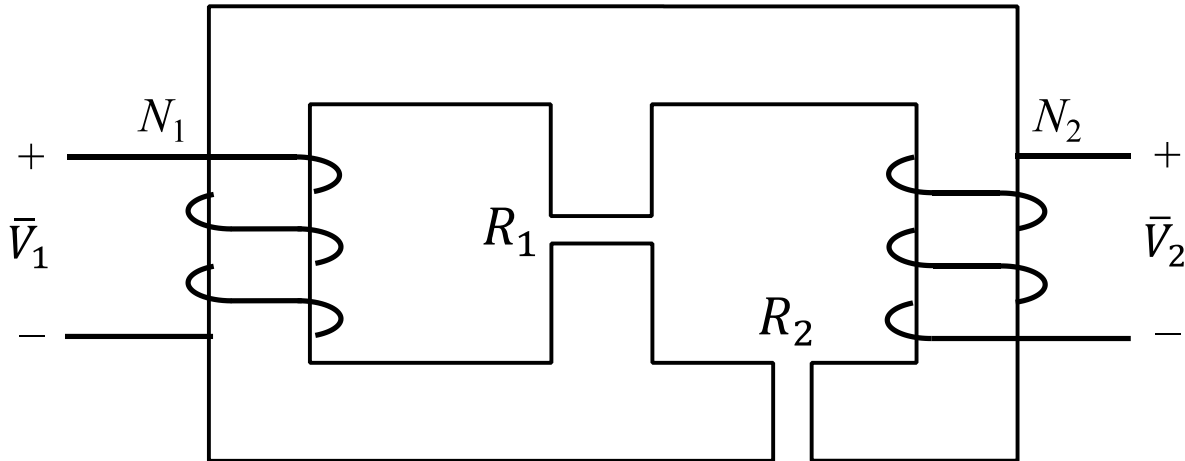
$\phi =$ _____

c) What is the voltage drop across the coil in terms of the unknown current i ? (5 points)

$v =$ _____

Problem 4 (25 Points)

You are consulting for a wind farm company who is looking to install a high voltage transformer to minimize their transmission loss. A schematic of the transformer is shown below.



The first coil has $N_1 = 10$ turns while the second coil has $N_2 = 500$ turns. The transformer contains two air gaps, with reluctances $R_1 = 50 \text{ At/Wb}$ and $R_2 = 1000 \text{ At/Wb}$ as labeled. You may treat the permeability of the magnetic core as infinite.

a) Assuming for now that the transformer is ideal, what would the output voltage $\bar{V}_2$ be (at open circuit) with an input voltage $\bar{V}_1 = 1\angle 0^\circ \text{ kV (rms)}$ at 60 Hz? (Please specify both the phase and the magnitude of the output voltage $\bar{V}_2$, and briefly justify your answer.)

$\bar{V}_2 =$ _____

Reason: _____

b) Compute the self inductances L_1 , L_2 , and the mutual inductance M of the two coils.

$L_1 =$ _____

$L_2 =$ _____

$M =$ _____

c) Use the self and mutual inductance values you have just computed above as a more realistic model of the transformer. What would the output voltage $\bar{V}_2$ be (at open circuit) with an input voltage $\bar{V}_1 = 1\angle 0^\circ$ kV (rms) at 60 Hz? (Please specify both the phase and the magnitude of the output voltage $\bar{V}_2$, and briefly justify your answer.)

$\bar{V}_2 =$ _____
Reason: _____

